

Homework 6

Differential Geometry

Due on June 7

Important. Only **handwritten solutions** will be accepted; typed solutions are **not** allowed. In all problems, you must show your reasoning and derivations clearly, step by step. Although it is easy to find answers using online tools, the main purpose of this homework is to deepen your understanding through careful, hands-on problem solving. Working through the details yourself is an essential part of learning the material.

Problem 1. Show that

$$\mathrm{SO}(4) \cong \{\mathrm{SU}(2) \times \mathrm{SU}(2)\} / \{\pm \mathrm{Id}\},$$

where $\mathrm{Id} \hookrightarrow \mathrm{SU}(2) \times \mathrm{SU}(2)$ is the diagonal embedding. The hint is given as follows.

Let $\mathbb{H}$ be the quaternions. An element $x \in \mathbb{H}$ can be expressed as

$$x = x_1 + x_2i + x_3j + x_4k,$$

where $x_a \in \mathbb{R}$ for $a = 1, \dots, 4$, and

$$i^2 = j^2 = k^2 = -1, \quad ij = -ji = k, \quad jk = -kj = i, \quad ki = -ik = j.$$

We define the imaginary part of x by

$$\mathrm{Im} x = x_2i + x_3j + x_4k,$$

so that the conjugate $\bar{x}$ is written as

$$\bar{x} = x_1 - x_2i - x_3j - x_4k.$$

Therefore,

$$\overline{xy} = \bar{y} \bar{x}.$$

The norm of x is

$$|x|^2 = x\bar{x} = \bar{x}x = x_1^2 + x_2^2 + x_3^2 + x_4^2.$$

From this viewpoint, $\mathrm{SU}(2)$ can be considered as the group of unit quaternions,

$$\mathrm{SU}(2) = \{x \in \mathbb{H} \mid |x| = 1\}.$$

Then $\mathrm{SU}(2) \times \mathrm{SU}(2)$ acts on $\mathbb{H}$ by rotations as follows:

$$x \longmapsto q_1 x q_2^{-1}$$

is a rotation of $\mathbb{R}^4 = \mathbb{H}$ for $q_1, q_2 \in \mathrm{SU}(2)$. Moreover, $(-q_1, -q_2)$ represents the same rotation as (q_1, q_2) . Show that these represent all rotations of $\mathbb{R}^4 = \mathbb{H}$, so that the quotient is isomorphic to $\mathrm{SO}(4)$.

Problem 2. Derive the fundamental groups of $\mathrm{SO}(2)$, $\mathrm{SO}(3)$, and $\mathrm{SO}(4)$.

Problem 3. We have seen that

$$\mathrm{PSL}(2, \mathbb{R}) = \mathrm{SL}(2, \mathbb{R}) / \{\pm \mathrm{Id}\}$$

acts on the upper half-plane

$$\mathbb{H} = \{z \in \mathbb{C} \mid \mathrm{Im} z > 0\}, \quad ds^2 = \frac{dz d\bar{z}}{(\mathrm{Im} z)^2},$$

as an isometry group:

$$\mathrm{PSL}(2, \mathbb{R}) \times \mathbb{H} \rightarrow \mathbb{H}, \quad \begin{pmatrix} a & b \\ c & d \end{pmatrix} \cdot z \mapsto \frac{az + b}{cz + d}.$$

Show that this action is transitive and find the stabilizer subgroup of the point $i \in \mathbb{H}$. Find the fundamental group of $\mathrm{PSL}(2, \mathbb{R})$.

Problem 4. Show that $\mathrm{SL}(2, \mathbb{R})$ is a deformation retract to $\mathrm{SO}(2)$ in the following way.

- Write an element $A \in \mathrm{SL}(2, \mathbb{R})$ as $A = (a_1, a_2)$, where a_i are column vectors. The Gram-Schmidt process replaces A by

$$U = \left(\frac{a_1}{|a_1|}, \frac{a_2 - \frac{\langle a_1, a_2 \rangle}{|a_1|^2} a_1}{\left| a_2 - \frac{\langle a_1, a_2 \rangle}{|a_1|^2} a_1 \right|} \right) \in \mathrm{SO}(2),$$

where $\langle a_1, a_2 \rangle$ is the standard inner product of $\mathbb{R}^2$. Construct a deformation retract that connects A to U inside $\mathrm{SL}(2, \mathbb{R})$.

Do the same exercise for $\mathrm{SL}(2, \mathbb{C})$. Which Lie group does it have a deformation retract to? Derive the homology groups and the fundamental group of $\mathrm{SL}(2, \mathbb{R})$ and $\mathrm{SL}(2, \mathbb{C})$.

Problem 5 (Hopf fibration and Dirac monopole). We define

$$S^{2n+1} = \left\{ (z_0, \dots, z_n) \in \mathbb{C}^{n+1} \mid \sum_{i=0}^n |z_i|^2 = 1 \right\},$$

and let the complex projective space $\mathbb{CP}^n$ be the quotient space $S^{2n+1}/\sim$, where

$$(z_0, \dots, z_n) \sim \lambda(z_0, \dots, z_n), \quad \lambda \in U(1).$$

We denote a point of $\mathbb{CP}^n$ by $[z_0; \dots; z_n]$.

When $n = 1$, we can consider S^3 as the principal $U(1)$ -bundle over $\mathbb{CP}^1$:

$$\begin{array}{ccc} S^1 & \longrightarrow & S^3 \\ & & \downarrow \pi \\ & & \mathbb{CP}^1 \end{array} .$$

First, show that $\mathbb{CP}^1$ can be identified with $S^2 \subset \mathbb{R}^3$ by the map

$$[z_0; z_1] \longmapsto \left(z_0 \bar{z}_1 + \bar{z}_0 z_1, -i(z_0 \bar{z}_1 - \bar{z}_0 z_1), |z_0|^2 - |z_1|^2 \right).$$

Let us define trivializations of this bundle over

$$U_N = S^2 \setminus \{\text{north pole}\}, \quad U_S = S^2 \setminus \{\text{south pole}\},$$

by

$$\begin{aligned} \pi^{-1}(U_N) &\rightarrow U_N \times S^1, & (z_0; z_1) &\longmapsto \left([z_0; z_1], \frac{z_0}{|z_0|} \right), \\ \pi^{-1}(U_S) &\rightarrow U_S \times S^1, & (z_0; z_1) &\longmapsto \left([z_0; z_1], \frac{z_1}{|z_1|} \right). \end{aligned}$$

Find the transition function between these trivializations.

Let us consider the embeddings

$$\begin{aligned} f_N: U_N &\hookrightarrow \pi^{-1}(U_N), & (\theta, \phi) &\longmapsto \left(\cos \frac{\theta}{2}, \sin \frac{\theta}{2} e^{-i\phi} \right), \\ f_S: U_S &\hookrightarrow \pi^{-1}(U_S), & (\theta, \phi) &\longmapsto \left(\cos \frac{\theta}{2} e^{i\phi}, \sin \frac{\theta}{2} \right), \end{aligned}$$

where (θ, ϕ) is the standard polar coordinate of S^2 . Defining a connection

$$A = i \operatorname{Im}(z_0 dz_0 + z_1 dz_1)$$

on $S^3 \subset \mathbb{C}^2$, show that

$$\begin{aligned} f_N^* A &= -\frac{i}{2}(1 - \cos \theta) d\phi && \text{on } U_N, \\ f_S^* A &= \frac{i}{2}(1 + \cos \theta) d\phi && \text{on } U_S. \end{aligned}$$

Show that they are related by gauge transformations and that the curvature F is well-defined on S^2 . This connection provides the **Dirac magnetic monopole**:

$$\frac{i}{2\pi} \int_{S^2} F = 1.$$